

Bank Management Database System

Project Phase 1 Report

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1 Project Description

This project involves the design and implementation of a comprehensive Bank Management Database System. The primary goal is to provide a structured framework for managing the complex operations of financial institutions. The system tracks the hierarchical structure of banks, their physical branches across various countries, and the workforce (managers and employees) that operates them. Additionally, it handles customer data, account management, and a variety of financial transactions, including specialized types like transfers and loans.

2 Entities

The system is built upon the following core entities:

- **Bank:** The overarching financial institution.
- **Country:** The geographical locations where banks operate.
- **Branch:** Specific physical locations belonging to a bank.
- **Manager/Employee:** The human resources governing and working within branches.
- **Customer:** Individuals using the bank's services.
- **Account:** Financial holdings owned by customers and hosted by branches.
- **Transaction:** Financial activities with specialized subtypes:
 - **Transfer:** Moving funds between specific accounts.
 - **Loan:** Credit extended to customers with a set maturity date.

3 Attributes

Each entity is defined by specific attributes (Primary Keys are denoted with underline):

- **Manager:** Manager ID, Full Name.
- **Employee:** Employee ID, Full Name, Position, Salary.
- **Branch:** Branch ID, Address.

- **Bank:** Bank ID, Name.
- **Country:** Country ID, Name.
- **Customer:** Customer ID, Full Name, Phone, Email.
- **Account:** Account ID, Number, Balance, Open (Status).
- **Transaction:** Transaction ID, Amount, Currency, Date.
 - *Transfer (Subtype)*: Recipient.
 - *Loan (Subtype)*: Maturity.

4 Relationships

The following relationships define the interaction between entities, incorporating specific relationship attributes where noted:

- **Manages:** A $1 : M$ relationship between Manager and Employee.
 - *Attributes*: From, Until (tracking the duration of management).
- **Works In:** A $M : 1$ relationship between Employee and Branch.
 - *Attributes*: From, Until (tracking employment duration at a specific branch).
- **Owned By:** A $M : 1$ relationship connecting Branch to its parent Bank.
- **Placed In:** A $M : M$ relationship between Bank and Country.
- **Owns:** A $1 : M$ relationship between Customer and Account.
- **Belongs To:** A $M : 1$ relationship connecting an Account to a specific Branch.
- **Make:** A $1 : M$ relationship between Account and Transaction records.

5 Constraints

To maintain the logic of the banking system, the following business rules and cardinality constraints are applied:

5.1 Domain Constraints

Domain constraints ensure that the data entered into each attribute falls within valid ranges and follows specific formats:

- **Salary:** Must be a positive numeric value strictly greater than 0 ($Salary > 0$).
- **Balance:** An account balance cannot be negative ($Balance \geq 0$).
- **Transaction Amount:** Every transaction must involve a positive value.

- **Phone Number:** The **Phone** attribute is restricted to numeric characters.
- **Mandatory Data (NOT NULL):** To ensure data integrity, all attributes are required and cannot be null, **with the sole exceptions of Customer Email and Customer Phone**. All other information, including Primary Keys, Names, Addresses, and Transaction details, must be provided.

5.2 Relationship Constraints

Based on the provided ER diagram, the following structural constraints are enforced to ensure data consistency:

- **Total Participation:** Represented by **bold lines** in the diagram. This indicates that every instance of an entity must participate in the relationship.
 - Every **Manager** must manage at least one **Employee**.
 - Every **Employee** must be assigned to a **Manager** and belong to a **Branch**.
 - Every **Branch** must be owned by a **Bank** and have at least one **Employee**.
 - Every **Bank** must operate in at least one **Country** and have at least one **Branch**.
 - Every **Account** must be owned by a **Customer** and belong to a **Branch**.
 - Every **Transaction** must be associated with an **Account**.
- **Constraints (Arrows):** Arrows indicate the "one" side of a relationship, while lines without arrows represent the "many" side.
 - **1:M (One-to-Many):** A single Manager manages multiple Employees; a Customer can own multiple Accounts; and an Account can have multiple Transactions.
 - **M:1 (Many-to-One):** Multiple Employees work at one Branch; multiple Branches are owned by one Bank; and multiple Accounts are hosted at one Branch.
 - **M:M (Many-to-Many):** Represented by the "Placed in" relationship without arrows on either side, allowing multiple Banks to operate in multiple Countries.
- **ISA:** The triangle symbol denotes that **Transfer** and **Loan** are specialized types of **Transactions**, inheriting its basic attributes while adding their own (*Recipient* and *Maturity*).

6 ER Diagram

The following diagram illustrates the logical structure of the database. It highlights the entities, their attributes, and the specific relationship constraints described in the previous sections.

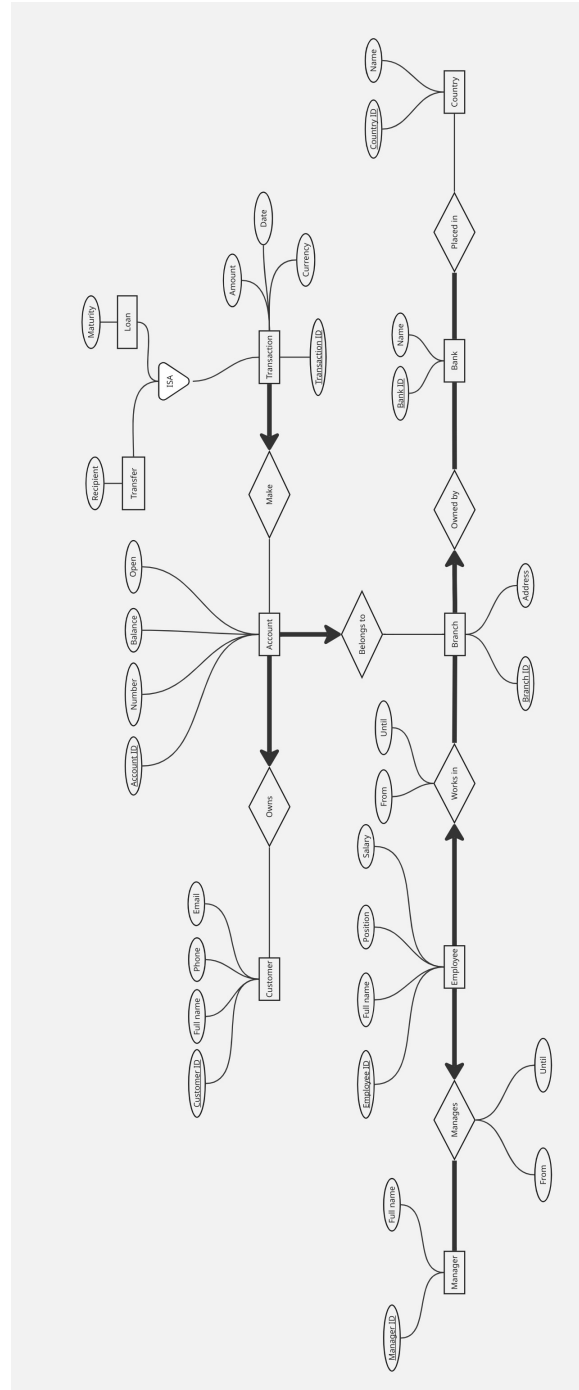


Figure 1: Entity-Relationship Diagram for the Bank Management System (Rotated)

7 Appendix

- **AI Tools Used:** Gemini (Google).
- **Scope of AI Usage:** AI was utilized strictly for the **wording, formatting, and structural organization** of the LaTeX report. While AI was occasionally consulted during the brainstorming phase for potential attributes and entities, the suggestions provided were not good enough. Consequently, the final selection of all entities, attributes, and logic was refined and written by the project members.
- **Originality Statement:** All core project ideas, the conceptual database design, and the logical framework of entities and relationships are entirely the original work of the team members.
- **Software and Tools:**
 - **Diagramming:** The Entity-Relationship Diagram was designed and exported using **Miro.com**.
 - **Report Writing:** The final documentation was composed using **LaTeX** within the **Overleaf** collaborative environment.
- **Human Verification:** All relationship cardinalities, participation constraints (bold lines), and primary keys were manually verified to ensure they accurately represent the diagram created on Miro.

8 Summary of AI Assistance and Transcript

This section provides a comprehensive overview of the interaction between the project team and the AI (Gemini) to ensure full transparency regarding the report’s generation.

8.1 Executive Summary of Collaboration

The primary role of the AI in this project was to act as a technical editor and LaTeX consultant. While the conceptual database design, entity selection, and relationship logic were developed entirely by the project team using **Miro.com**, the AI assisted in:

- Translating raw project descriptions into a structured, academic LaTeX format.
- Solving complex document positioning issues, such as image ”floating” and stretching.
- Standardizing technical terminology for relationship cardinalities ($1 : M$, $M : M$) and constraints.

8.2 Interaction Transcript

The following table summarizes the step-by-step assistance provided during the chat session:

User Request	AI Action / Implementation
Initial Report Setup	Converted a single-block text description into organized LaTeX sections (Entities, Attributes, etc.).
Cardinality Definition	Clarified the use of $1 : M$ and $M : M$ notation to match standard academic database modeling.
Image Formatting	Applied 90-degree rotation and <code>keepaspectratio</code> to ensure the Miro diagram fit the A4 page without distortion.
Layout Debugging	Integrated the <code>float</code> package and <code>[H]</code> specifier to fix the diagram’s position under the correct header.
Constraint Expansion	Drafted a detailed explanation of ”Total Participation” (bold lines) and ”Arrows” based on the team’s visual model.
Appendix & Summary	Generated the final documentation regarding tool usage (Miro/Overleaf) and a summary of the AI’s role.

Table 1: Chronological Log of AI Assistance

8.3 Statement of Originality

The project team affirms that all ”creative” aspects of the database—including the choice of entities, the specific attributes assigned to them, and the business logic governing the bank’s operations—were conceived by the team members. AI suggestions regarding the database schema were found insufficient for the project’s specific needs and were replaced by the team’s own refined logic. The AI’s contribution was limited to the **presentation layer** and **technical formatting** of this report including this section.